

# Labour Management System

How it works, with one worked example · SEPL ERP · August 2026

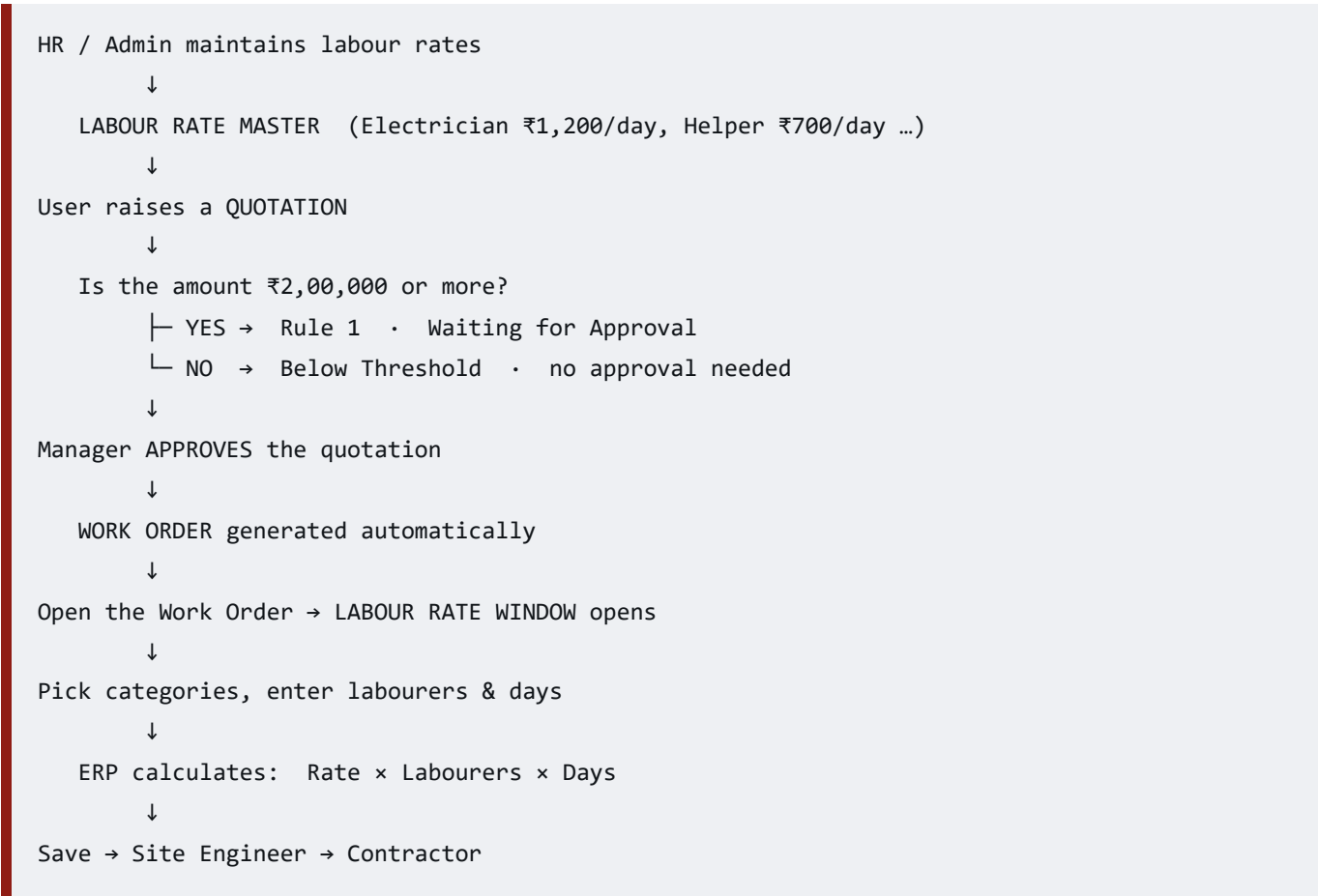
## Where it sits

A new sidebar group, **Labour Management System**, between CRM and Solar Division. It holds two pages:

| Page                   | What it does                                                                  |
|------------------------|-------------------------------------------------------------------------------|
| Quotations & Rates     | Dashboard, Quotations, and the Labour Rate Master that HR maintains           |
| Projects & Work Orders | Your existing Indent Labour Payment page, moved here. Same data, same access. |

**Nothing was duplicated.** Work Orders, muster roll, measurement book and contractor bills already existed in Indent Labour Payment. That module became the Labour Management System rather than a second one being built alongside it, so there is still only one place that owns a Work Order.

## The workflow



## Worked example: ABC Hotel electrical fit-out

### Step 1 — HR sets the rates (done once, then maintained)

HR opens **Quotations & Rates** → **Labour Rate Master** and adds the company rates. Nobody else can change these.

| Labour Category | Trade      | Dept | Unit | Standard Rate | Overtime | Status |
|-----------------|------------|------|------|---------------|----------|--------|
| Electrician     | Electrical | ELE  | Day  | ₹1,200        | ₹150     | Active |
| Helper          | Electrical | ELE  | Day  | ₹700          | ₹90      | Active |
| Supervisor      | Electrical | ELE  | Day  | ₹2,000        | ₹250     | Active |

### Step 2 — Raise the quotation

On the **Quotations** tab, click *New Quotation*. Client, contractor and project come from dropdowns fed by CRM, Procurement and Projects — nothing is re-typed.

| Field           | Value                          | Comes from                    |
|-----------------|--------------------------------|-------------------------------|
| Project         | ABC Hotel — Electrical Fit-out | Projects module               |
| Client          | ABC Hotels Pvt Ltd             | CRM → Customers               |
| Contractor      | XYZ Electricals                | Procurement → Sub-Contractors |
| Labour Category | Electrician                    | Labour Rate Master            |
| Amount          | ₹6,00,000                      | typed by the user             |

### Step 3 — The threshold rule fires automatically

As soon as the amount is entered, the ERP decides whether approval is needed. The user does nothing.

| Quotation | Amount    | Threshold | Rule   | Status               |
|-----------|-----------|-----------|--------|----------------------|
| QTN0001   | ₹6,00,000 | ₹2,00,000 | Rule 1 | Waiting for Approval |
| QTN0002   | ₹1,50,000 | ₹2,00,000 | —      | Below Threshold      |

₹6,00,000 is at or above ₹2,00,000, so it waits for approval. ₹1,50,000 is below it and clears without approval.

### Step 4 — Manager approves → Work Order appears by itself

The manager clicks the tick on QTN0001. The ERP creates **WO0001** against the project in the same step, carrying the contractor and the approved value across.

**Two safeguards here.** The person who raised the quotation cannot approve their own — someone else has to. And the quotation and its Work Order are created together, so you can never end up with an approved quotation that has no Work Order.

### Step 5 — The Labour Rate Window opens

Opening WO0001 and clicking *Labour Rate Window* shows every active HR rate, with search and filters for Department, Trade, Category and Skill Level. The user enters only **how many labourers** and **how many days** — the rate column is read-only.

**Step 6 — The ERP calculates the cost**

$$\text{Total} = \text{Rate} \times \text{Labourers} \times \text{Days}$$

| Category    | Rate   | Labourers | Days | Total     |
|-------------|--------|-----------|------|-----------|
| Electrician | ₹1,200 | 5         | 10   | ₹60,000   |
| Helper      | ₹700   | 8         | 10   | ₹56,000   |
| Supervisor  | ₹2,000 | 1         | 10   | ₹20,000   |
| Grand Total |        |           |      | ₹1,36,000 |

The Labour Cost Summary card on the Work Order shows:

| Categories | Total labourers | Man-days | Avg per category | Total labour cost |
|------------|-----------------|----------|------------------|-------------------|
| 3          | 14              | 140      | ₹45,333          | ₹1,36,000         |

**Why man-days, not days.** The three crews all worked the same 10 calendar days. Adding the days column would give 30, which means nothing. 14 labourers × 10 days = **140 man-days** is the number that actually drives the cost.

## The rules that protect you

### 1. Nobody except HR can change a rate

The rate column is read-only in the window, and the server re-reads the rate from the Labour Rate Master when the line is saved. A user who tries to send their own rate straight to the system is ignored.

**Tested:** a request was sent with a rate of ₹99,999 attached. The system saved **₹1,200** — the HR rate. The read-only-ness is real, not just a greyed-out box on screen.

### 2. An existing Work Order keeps the rate it was created with

If HR raises the Electrician rate next month, WO0001 does not silently get more expensive.

| Event                         | Electrician rate | WO0001 total                 | A new WO would be         |
|-------------------------------|------------------|------------------------------|---------------------------|
| Work Order created today      | ₹1,200           | ₹1,36,000                    | —                         |
| HR revises the rate to ₹1,500 | ₹1,500           | <b>₹1,36,000</b> (unchanged) | ₹75,000 for the same crew |

### 3. Every rate change is recorded, with a reason

A rate cannot be changed without saying why. The system keeps who changed it, the old rate, the new rate, the reason, and the date and time.

| When        | Category    | Action  | Old    | New    | Reason               | By    |
|-------------|-------------|---------|--------|--------|----------------------|-------|
| 08 Aug 2026 | Electrician | updated | ₹1,200 | ₹1,500 | Annual wage revision | Admin |
| 01 Aug 2026 | Electrician | created | —      | ₹1,200 | —                    | Admin |

### 4. Only one active rate per category and date

Two active Electrician rates for the same trade on the same effective date are refused by the database itself, not just by a screen check — so two people saving at the same moment cannot slip a duplicate through.

### 5. Everyone is told when rates move

*"Labour rates have been updated by HR. All new Work Orders will use the latest rates. Work Orders already created keep the rates they were created with."*

## Who can do what

| Role            | Labour Rate Master         | Quotations                   |
|-----------------|----------------------------|------------------------------|
| HR / Admin      | Create, edit, delete rates | Full access                  |
| Project Manager | Read only                  | Raise & approve (if granted) |
| Site Engineer   | Read only                  | Raise only                   |
| Contractor      | Read only                  | Read only                    |

|               |           |           |
|---------------|-----------|-----------|
| Everyone else | Read only | Read only |
|---------------|-----------|-----------|

Site engineers and contractors do not need a separate grant to see rates — anyone who works on Work Orders can open the Labour Rate Window automatically.

### To start using it

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- **Grant permissions** in Admin → Roles & Permissions. Two new rows: *Labour Quotations* (the `approve` tick is what generates the Work Order) and *Labour Rate Master* (give HR create/edit/delete).
- **HR adds the rates** on Quotations & Rates → Labour Rate Master. Until at least one active rate exists, the Labour Rate Window will be empty.
- **Raise the first quotation** and follow the six steps above.

### Not built yet

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This is the first slice — quotation through to labour cost. Still to come from the full specification:

- Labour master with attendance and wage register
- The six-stage bill verification workflow (Site Engineer → Site Head → Project Manager → Finance → Accounts → Payment)
- Payments screen and vendor ledger posting
- The remaining reports, and PDF export from inside the ERP (CSV export works today)

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SEPL ERP — Labour Management System. Figures shown are from a verified test run, not illustrations. The approval threshold of ₹2,00,000 is configurable without a code change.